Experiment 2

Aim:- Convert the ER diagram into a relational schema. Write complete MySQL CREATE TABLE statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints. Insert sample data and demonstrate referential integrity violations.

Source Code:-

1. Relational Schema

Table	Primary Key	Important Foreign Keys
CUSTOMER	CustomerID	—
CUSTOMER_PHONE	(CustomerID, Phone)	CustomerID → CUSTOMER
CUSTOMER_ALT_PHONE	(CustomerID, AltPhone)	CustomerID → CUSTOMER
SELLER	SellerID	—
SELLER_PHONE	(SellerID, Phone)	SellerID → SELLER
ADDRESS	AddressID	CustomerID → CUSTOMER
ADDRESS_MOBILE	(AddressID, MobileNo)	AddressID → ADDRESS
CATEGORY	CategoryID	ParentCategoryID → CATEGORY
PRODUCT	ProductID	SellerID → SELLER, CategoryID → CATEGORY
PRODUCT_IMAGE	(ProductID, ImageURL)	ProductID → PRODUCT
ELECTRONICS	ProductID	ProductID → PRODUCT
CLOTHING	ProductID	ProductID → PRODUCT
BOOKS	ProductID	ProductID → PRODUCT
ORDERS	OrderID	CustomerID → CUSTOMER, AddressID → ADDRESS
ORDER_COUPON	(OrderID, CouponCode)	OrderID → ORDERS
ORDERITEM	(OrderID, ItemNo)	OrderID → ORDERS, ProductID → PRODUCT
PAYMENT	PaymentID	OrderID → ORDERS
DELIVERY	DeliveryID	OrderID → ORDERS, AddressID → ADDRESS

2. SQL Queries

```
CREATE DATABASE ECOMMERCE;
USE ECOMMERCE;

CREATE TABLE
 CUSTOMER( CustomerID INT
 PRIMARY KEY, Name
 VARCHAR(50) NOT NULL,
 Email VARCHAR(100) NOT NULL UNIQUE,
 Password VARCHAR(100) NOT NULL,
 DateOfBirth DATE NOT NULL
);

CREATE TABLE CUSTOMER_PHONE(
 CustomerID INT,
 Phone VARCHAR(15),
 PRIMARY KEY(CustomerID,Phone),
 FOREIGN KEY(CustomerID) REFERENCES CUSTOMER(CustomerID)
 ON DELETE CASCADE
);

CREATE TABLE CUSTOMER_ALT_PHONE(
 CustomerID INT,
 AltPhone VARCHAR(15),
 PRIMARY KEY(CustomerID,AltPhone),
 FOREIGN KEY(CustomerID) REFERENCES CUSTOMER(CustomerID)
 ON DELETE CASCADE
);

CREATE TABLE
 SELLER( SellerID INT PRIMARY
 KEY,
 SellerName VARCHAR(50) NOT NULL,
 Email VARCHAR(100) UNIQUE NOT NULL,
 GSTIN VARCHAR(20) UNIQUE NOT NULL,
 AccountNo VARCHAR(20),
 IFSCCode VARCHAR(15),
 BankName VARCHAR(50)
);

CREATE TABLE SELLER_PHONE(
 SellerID INT,
 Phone VARCHAR(15),
 PRIMARY KEY(SellerID,Phone),
 FOREIGN KEY(SellerID) REFERENCES SELLER(SellerID)
 ON DELETE CASCADE
);
```

```
CREATE TABLE
 CATEGORY( CategoryID INT
 PRIMARY KEY,
 CategoryName VARCHAR(50) NOT NULL,
 Description VARCHAR(100),
 ParentCategoryID INT,
 FOREIGN KEY(ParentCategoryID) REFERENCES CATEGORY(CategoryID)
 ON DELETE SET NULL
);

CREATE TABLE
 PRODUCT( ProductID INT
 PRIMARY KEY,
 ProductName VARCHAR(50) NOT NULL,
 Description VARCHAR(100),
 Price DECIMAL(10,2) NOT NULL,
 StockQty INT NOT NULL,
 Weight DECIMAL(8,2),
 SellerID INT NOT NULL,
 CategoryID INT,
 FOREIGN KEY(SellerID) REFERENCES SELLER(SellerID)
 ON DELETE CASCADE,
 FOREIGN KEY(CategoryID) REFERENCES CATEGORY(CategoryID)
 ON DELETE SET NULL
);

CREATE TABLE PRODUCT_IMAGE(
 ProductID INT,
 Image VARCHAR(200),
 PRIMARY KEY(ProductID,Image),
 FOREIGN KEY(ProductID) REFERENCES PRODUCT(ProductID)
 ON DELETE CASCADE
);

CREATE TABLE
 ELECTRONICS( ProductID INT
 PRIMARY KEY, Brand
 VARCHAR(50),
 Warranty VARCHAR(30),
 PowerRating VARCHAR(30),
 FOREIGN KEY(ProductID) REFERENCES PRODUCT(ProductID)
 ON DELETE CASCADE
);
```

```
CREATE TABLE
 CLOTHING( ProductID INT
 PRIMARY KEY, Size
 VARCHAR(20),
 Colour VARCHAR(20),
 Material VARCHAR(30),
 FOREIGN KEY(ProductID) REFERENCES PRODUCT(ProductID)
 ON DELETE CASCADE
);

CREATE TABLE
 BOOKS( ProductID INT PRIMARY
 KEY, ISBN VARCHAR(20)
 UNIQUE, Author VARCHAR(50),
 Publisher VARCHAR(50),
 FOREIGN KEY(ProductID) REFERENCES PRODUCT(ProductID)
 ON DELETE CASCADE
);

CREATE TABLE
 ADDRESS( AddressID INT
 PRIMARY KEY, DoorNo
 VARCHAR(10),
 Street VARCHAR(50),
 Area VARCHAR(50),
 City VARCHAR(30),
 State VARCHAR(30),
 Pincode VARCHAR(10) NOT NULL,
 AddressType VARCHAR(20),
 CustomerID INT NOT NULL,
 FOREIGN KEY(CustomerID) REFERENCES CUSTOMER(CustomerID)
 ON DELETE CASCADE
);

CREATE TABLE ADDRESS_MOBILE(
 AddressID INT,
 MobileNo VARCHAR(15),
 PRIMARY KEY(AddressID,MobileNo),
 FOREIGN KEY(AddressID) REFERENCES ADDRESS(AddressID)
 ON DELETE CASCADE
);
```

```
CREATE TABLE
 ORDERS( OrderID INT
 PRIMARY KEY,
 OrderDate DATE NOT NULL,
 OrderStatus VARCHAR(20) NOT NULL,
 TotalAmount DECIMAL(10,2) NOT NULL,
 CustomerID INT NOT NULL,
 AddressID INT,
 FOREIGN KEY(CustomerID) REFERENCES CUSTOMER(CustomerID)
 ON DELETE CASCADE,
 FOREIGN KEY(AddressID) REFERENCES ADDRESS(AddressID)
 ON DELETE SET NULL
);

CREATE TABLE ORDER_COUPON(
 OrderID INT,
 CouponCode VARCHAR(30),
 PRIMARY KEY(OrderID,CouponCode),
 FOREIGN KEY(OrderID) REFERENCES ORDERS(OrderID)
 ON DELETE CASCADE
);

CREATE TABLE ORDERITEM(
 OrderID INT,
 ItemNo INT,
 ProductID INT NOT NULL,
 ItemPrice DECIMAL(10,2) NOT NULL,
 Quantity INT NOT NULL,
 Discount DECIMAL(10,2),
 TaxAmount DECIMAL(10,2),
 PRIMARY KEY(OrderID,ItemNo),
 FOREIGN KEY(OrderID) REFERENCES ORDERS(OrderID)
 ON DELETE CASCADE,
 FOREIGN KEY(ProductID) REFERENCES PRODUCT(ProductID)
 ON DELETE CASCADE
);

CREATE TABLE
 PAYMENT( PaymentID INT
 PRIMARY KEY,
 PaymentDate DATE NOT NULL,
 Amount DECIMAL(10,2) NOT NULL,
 PaymentMethod VARCHAR(30) NOT NULL,
 TransactionID VARCHAR(30) UNIQUE,
 PaymentStatus VARCHAR(20) NOT NULL,
 OrderID INT UNIQUE NOT NULL,
 FOREIGN KEY(OrderID) REFERENCES ORDERS(OrderID)
 ON DELETE CASCADE
);
```

```
CREATE TABLE
 DELIVERY( DeliveryID INT
 PRIMARY KEY,
 DeliveryDate DATE,
 DeliveryStatus VARCHAR(20) NOT NULL,
 TrackingID VARCHAR(30) UNIQUE,
 DeliveryPartner VARCHAR(50),
 ShippingCharge DECIMAL(8,2),
 OrderID INT UNIQUE NOT NULL,
 AddressID INT,
 FOREIGN KEY(OrderID) REFERENCES ORDERS(OrderID)
 ON DELETE CASCADE,
 FOREIGN KEY(AddressID) REFERENCES ADDRESS(AddressID)
 ON DELETE SET NULL
);

INSERT INTO CUSTOMER VALUES
(1,'Yeshshsvi','yujii@gmail.com','pass123','2006-06-10');

INSERT INTO CUSTOMER_PHONE VALUES
(1,'9214293443');

INSERT INTO CUSTOMER_ALT_PHONE VALUES
(1,'9123456780');

INSERT INTO SELLER VALUES
(101,'TechStore','tech@gmail.com','GST123',
'123456','SBIN001','SBI');

INSERT INTO SELLER_PHONE VALUES
(101,'9123456789');

INSERT INTO CATEGORY VALUES
(201,'Electronics','Electronic items',NULL);

INSERT INTO CATEGORY VALUES
(202,'Mobiles','Mobile phones',201);

INSERT INTO PRODUCT VALUES
(301,'Laptop','Gaming Laptop',75000,10,2.5,101,201);

INSERT INTO ELECTRONICS VALUES
(301,'ASUS','2 Years','180W');

INSERT INTO PRODUCT_IMAGE VALUES
(301,'laptop.jpg');

INSERT INTO ADDRESS VALUES
(401,'12','Eldeco','Transport Nagar','Lucknow',
'Uttar Pradesh','160017','Home',1);
```

```
INSERT INTO ADDRESS_MOBILE VALUES
(401,'9876543210');

INSERT INTO ORDERS VALUES
(501,'2026-08-14','Placed',75000,1,401);

INSERT INTO ORDERITEM VALUES
(501,1,301,75000,1,1000,9000);

INSERT INTO ORDER_COUPON VALUES
(501,'WELCOME10');


INSERT INTO PAYMENT VALUES
(601,'2026-08-14',74000,'UPI',
'TXN123','Paid',501);

INSERT INTO DELIVERY VALUES
(701,NULL,'Processing','TRK123',
'Delhivery',100,501,401);

-- REFERENTIAL INTEGRITY VIOLATION
INSERT INTO ORDERS VALUES
(502,'2026-08-14','Placed',50000,999,401);

-- SET NULL DEMONSTRATION
DELETE FROM CATEGORY WHERE CategoryID=201;

-- CASCADE DEMONSTRATION
DELETE FROM CUSTOMER WHERE CustomerID=1;
```

Output:-

OUTPUT 1 — SHOW TABLES

After creating all tables:

SHOW TABLES;

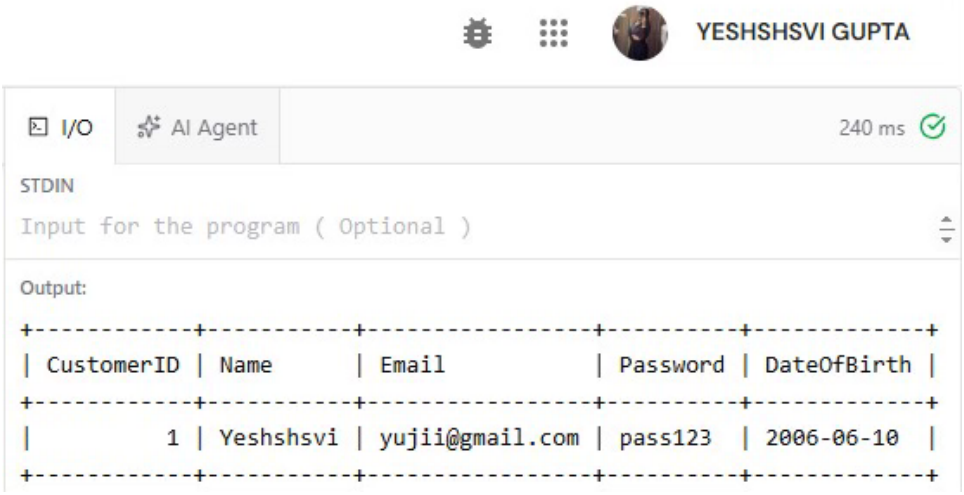


OUTPUT 2 — Sample data

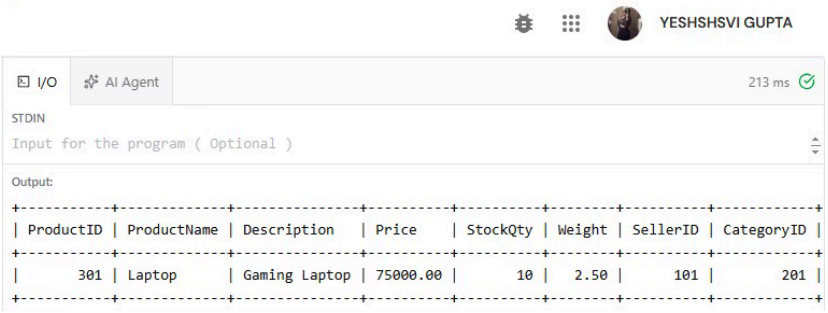
Use:

SELECT * FROM CUSTOMER;
SELECT * FROM PRODUCT;
SELECT * FROM ORDERS;

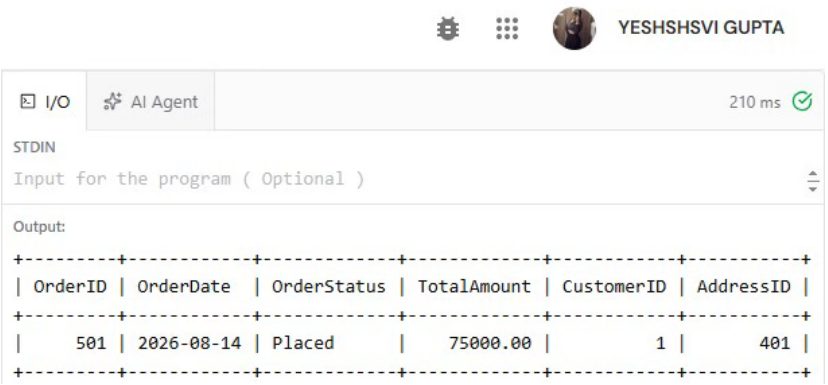
CUSTOMER



PRODUCT



ORDERS



OUTPUT 3 — Referential integrity violation

INSERT INTO ORDERS VALUES
(502,'2026-08-14','Placed',50000,999,401);

Since CustomerID = 999 doesn't exist:

```
Output:
ERROR 1452 (23000) at line 245: Cannot add or update a child row:
```

Reason: CustomerID 999 does not exist in CUSTOMER.

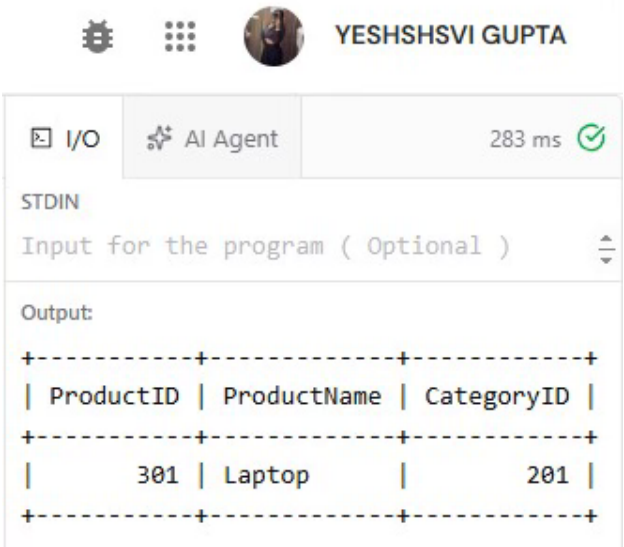
Satisfies "demonstrate referential integrity violations."

OUTPUT 4 — ON DELETE SET NULL

AIM explicitly asks us to use SET NULL.

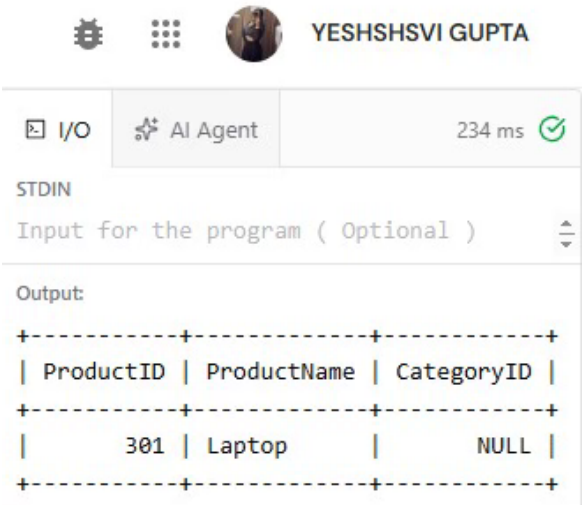
Before deletion:

SELECT ProductID, ProductName, CategoryID
FROM PRODUCT;



DELETE FROM CATEGORY WHERE CategoryID=201;

SELECT ProductID, ProductName, CategoryID
FROM PRODUCT;



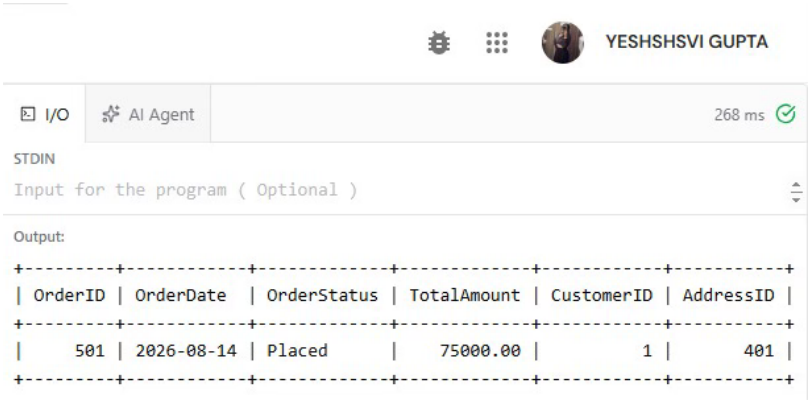
ON DELETE SET NULL demonstrated successfully.

OUTPUT 5 — ON DELETE CASCADE

Before deleting:

SELECT * FROM ORDERS;

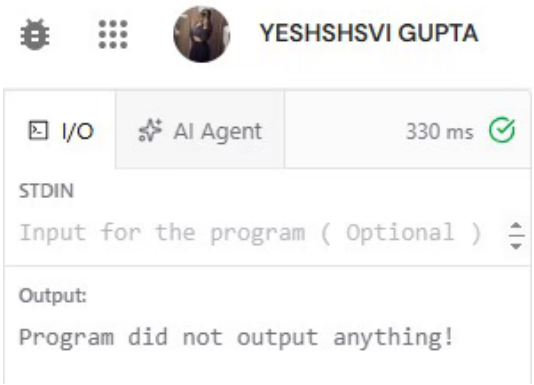
Output:



DELETE FROM CUSTOMER WHERE CustomerID=1;

SELECT * FROM ORDERS;

Output:



SELECT * FROM CUSTOMER_PHONE;

Output:

